



Trustworthy AI

Indrė Žliobaitė

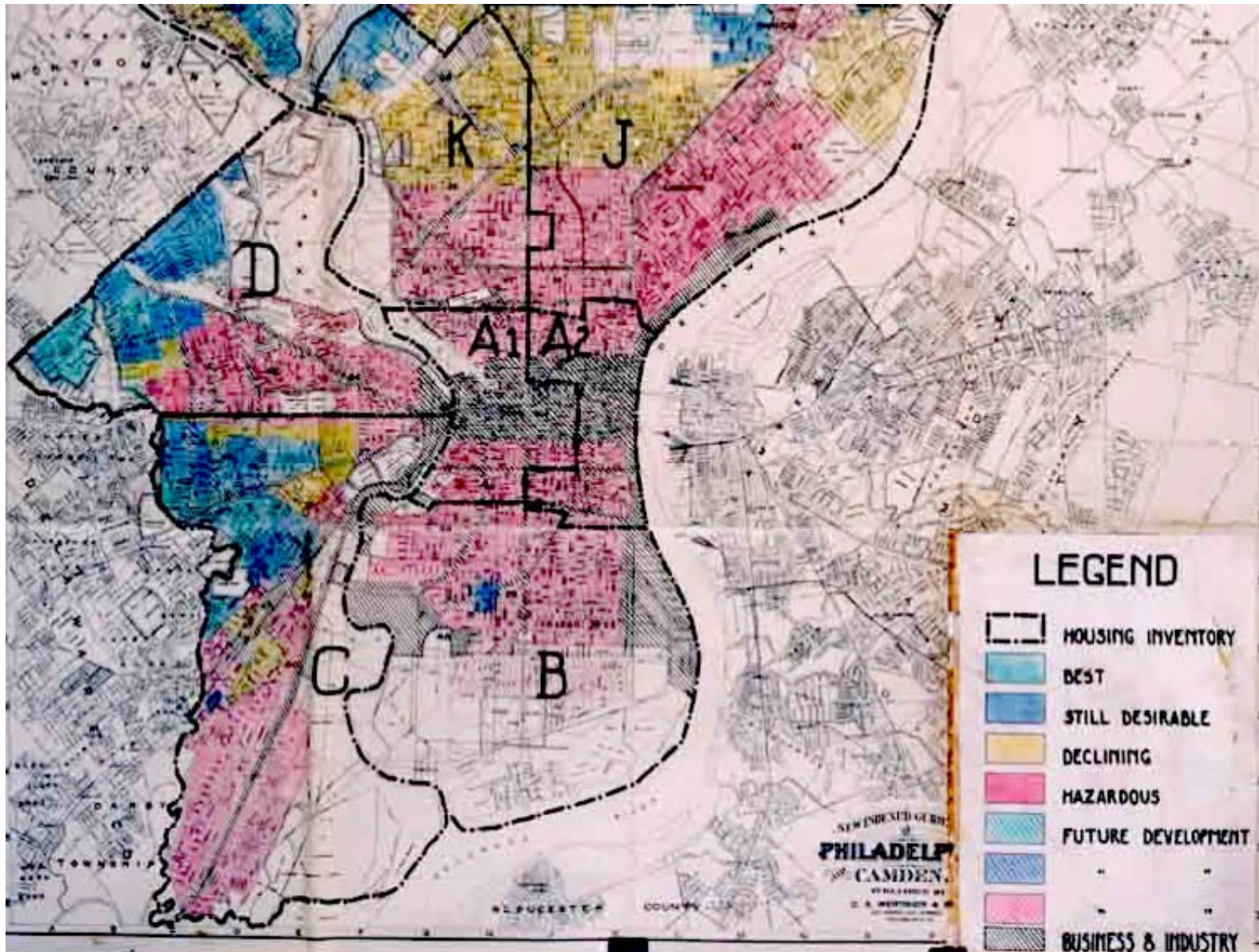
Professor

University of Helsinki

July 8, 2026

Vienna

Redlining



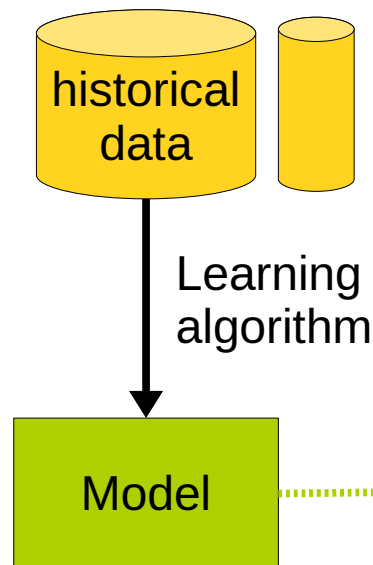
Source: "Home Owners' Loan Corporation Philadelphia redlining map". Licensed under Public Domain via Wikipedia

If there are biases in the data on which a model is trained

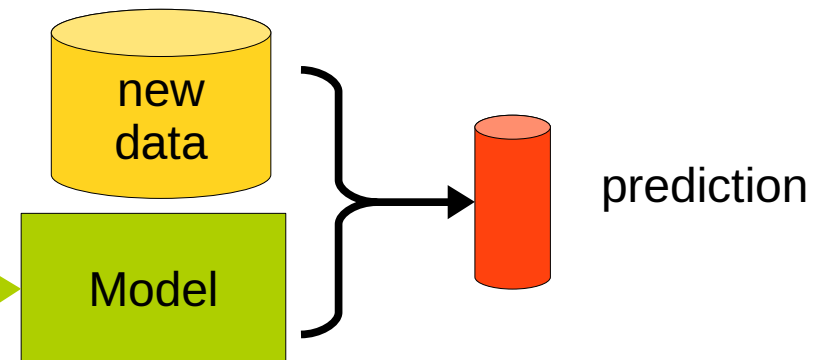
Unless instructed otherwise, the learned model will carry those biases forward

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?



Machine learning:
model is a summary of data



Measuring

No discrimination

	medicine		computer	
	female	male	female	male
Number of applicants	800	200	200	800
Acceptance rate	20%	20%	40%	40%
Accepted (+)	160	40	80	320

Measuring

No discrimination

	medicine		computer	
	female	male	female	male
Number of applicants	800	200	200	800
Acceptance rate	20%	20%	40%	40%
Accepted (+)	160	40	80	320

Discrimination is present

	medicine		computer	
	female	male	female	male
Number of applicants	800	200	200	800
Acceptance rate	15%	25%	35%	45%
Accepted (+)	120	50	70	360

How to correct?

What should be the acceptance rate?

Measuring

No discrimination

	medicine		computer	
	female	male	female	male
Number of applicants	800	200	200	800
Acceptance rate	20%	20%	40%	40%
Accepted (+)	160	40	80	320

Is there discrimination, or not?

	medicine		computer	
	female	male	female	male
Number of applicants	800	200	200	800
Acceptance rate	5%	5%	55%	55%
Accepted (+)	40	10	110	440

Redlining / indirect discrimination

Is there discrimination, or not?

	medicine		computer	
	female	male	female	male
Number of applicants	800	200	200	800
Acceptance rate	5%	5%	55%	55%
Accepted (+)	40	10	110	440

2000
30%
600

Redlining / indirect discrimination

	medicine		computer		
	female	male	female	male	
Number of applicants	800	200	200	800	2000
Acceptance rate	20%	20%	40%	40%	30%
Accepted (+)	160	40	80	320	600
	200		400		

	medicine		computer		
	female	male	female	male	
Number of applicants	800	200	200	800	2000
Acceptance rate	5%	5%	55%	55%	30%
Accepted (+)	40	10	110	440	600
	50		550		

New data, is there discrimination or not?

Is there discrimination, or not? Redlining?

	postal code I		postal code II	
	immigrant	native	immigrant	native
Number of applicants	800	200	200	800
Acceptance rate	5%	5%	55%	55%
Accepted (+)	40	10	110	440

Which is the “right” level?

Table 8.1 Summary statistics of the Adult dataset

	hours per week	annual income (K\$)	
female	36.4	5.8\$/h	10.9
male	42.4	13.8\$/h	30.4
all data	40.4	11.4\$/h	23.9

← Like male?

← Like all?

Redistribution of the same total resources?

Sufficiency

(equalized odds?)

$P(\text{positive outcome} \mid \text{male}, \text{positive decision})$

=

$P(\text{positive outcome} \mid \text{female}, \text{positive decision})$

?

Separation

(equal opportunity?)

$P(\text{positive decision} \mid \text{male}, \text{positive outcome})$

=

$P(\text{positive decision} \mid \text{female}, \text{positive outcome})$

?

This online textbook is an incomplete work in progress. Essential chapters are still missing. In the spirit of open review, we solicit broad feedback that will influence existing chapters, as well as the development of later material.

Table 3: Non-discrimination criteria

Independence	Separation	Sufficiency
$R \perp A$	$R \perp A \mid Y$	$Y \perp A \mid R$

Table 6: List of demographic fairness criteria

Name	Closest relative	Note	Reference
Statistical parity	Independence	Equivalent	Dwork et al. (2011)
Group fairness	Independence	Equivalent	
Demographic parity	Independence	Equivalent	
Conditional statistical parity	Independence	Relaxation	Corbett-Davies et al. (2017) Zliobaite et al 2011
Darlington criterion (4)	Independence	Equivalent	Darlington (1971)
Equal opportunity	Separation	Relaxation	Hardt, Price, Srebro (2016)
Equalized odds	Separation	Equivalent	Hardt, Price, Srebro (2016)
Conditional procedure accuracy	Separation	Equivalent	Berk et al. (2017)
Avoiding disparate mistreatment	Separation	Equivalent	Zafar et al. (2017)
Balance for the negative class	Separation	Relaxation	Kleinberg, Mullainathan, Raghavan (2016)
Balance for the positive class	Separation	Relaxation	Kleinberg, Mullainathan, Raghavan (2016)
Predictive equality	Separation	Relaxation	Chouldechova (2016)
Equalized correlations	Separation	Relaxation	Woodworth (2017)
Darlington criterion (3)	Separation	Relaxation	Darlington (1971)
Cleary model	Sufficiency	Equivalent	Cleary (1966)
Conditional use accuracy	Sufficiency	Equivalent	Berk et al. (2017)
Predictive parity	Sufficiency	Relaxation	Chouldechova (2016)
Calibration within groups	Sufficiency	Equivalent	Chouldechova (2016)
Darlington criterion (1), (2)	Sufficiency	Relaxation	Darlington (1971)

Confusion matrix

		True outcome	
		Positive	Negative
Predicted class	Positive	TP true positives	FP false positives
	Negative	FN false negatives	TN true negatives

$$N = FP + TP + FN + TN$$

(total number of applicants)

More measures

Measure	Statistic	Basis for use
<i>Measures on positively predicted cases</i>		
Statistical parity	$(FP + TP)/N$	Disparate impact law; 80% rule [7, 38]; [35, 52]
Predictive Parity, Precision	$TP/(FP + TP)$	Information retrieval [59]; clinical [2]; [21]
False Discovery Rate	$FP/(FP + TP)$	Multiple testing [10]; information retrieval [59]; [21, 26]
<i>Measures on negatives</i>		
Predictive Equality, False Positive Rate	$FP/(FP + TN)$	COMPAS audit [5, 21]; harm to innocents [25]
True Negative Rate, Specificity	$TN/(FP + TN)$	Clinical diagnostics, screening [1]; [43]
False Omission Rate	$FN/(FN + TN)$	[33]
Treatment Equality	FN/FP	[11]; not expressable as probability
<i>Measures on true outcomes</i>		
Equal Opportunity, True Positive Rate	$TP/(TP + FN)$	Recall in ML [59]; benefit to qualified [35, 43]
False Negative Rate	$FN/(TP + FN)$	Clinical screening [1]; wrongful denial [7, 43]
Equalised Odds, Separation	TPR & FPR	[6, 43]
Accuracy	$(TP + TN)/N$	Standard in ML
<i>Other measures</i>		
Calibration / Sufficiency	$P(y = 1 \mid f, a)$	Clinical prediction [75]; ML fairness [6, 21, 54]
FP Group	FP/N	City of Amsterdam custom [15]

More measures

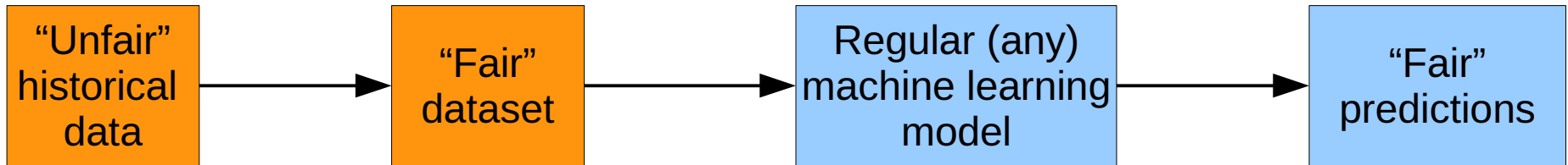
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Fairness-aware machine learning algorithmic principles

work with a selected fairness measure
that depends on selected sensitive attributes

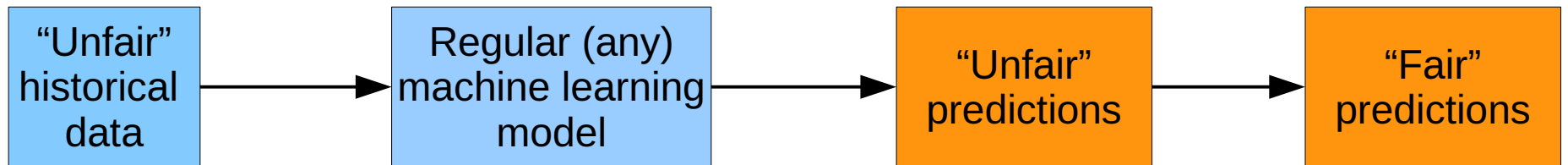
Algorithmic techniques

- Pre-processing



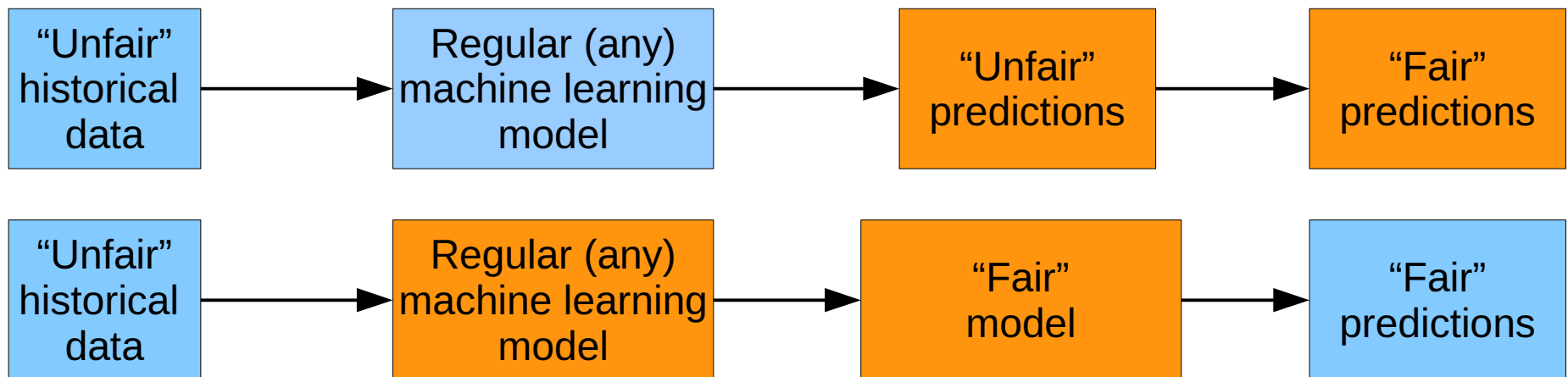
Algorithmic techniques

- Post-processing



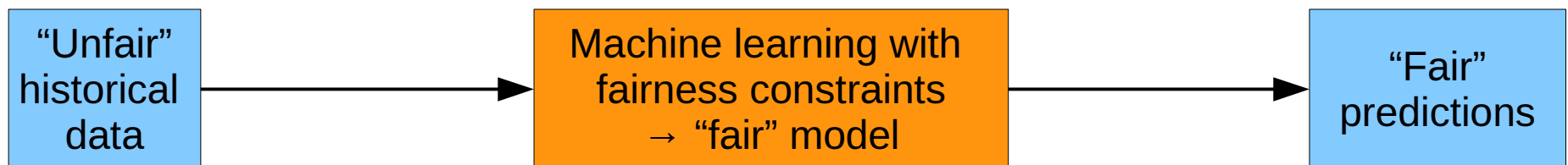
Algorithmic techniques

- Post-processing



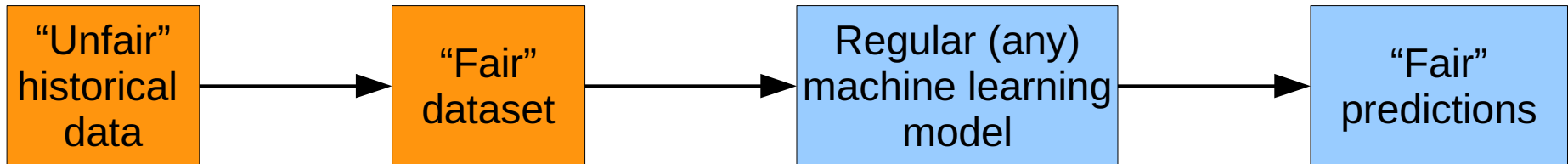
Algorithmic techniques

- Learning with fairness constraints

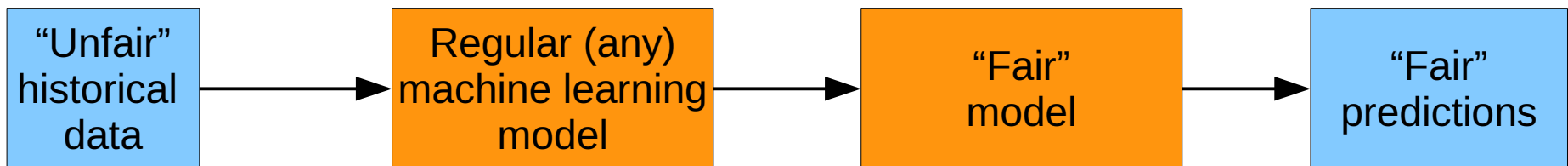
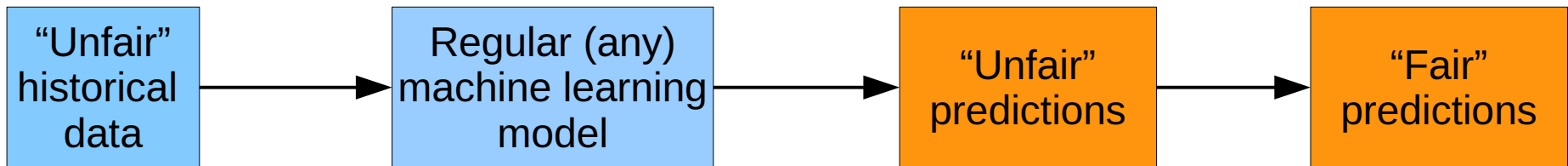


Algorithmic techniques

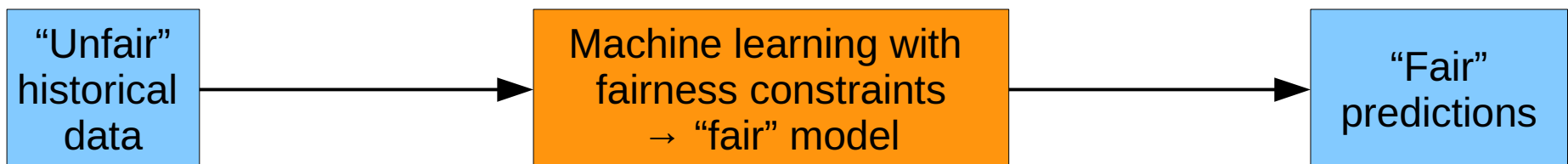
- Pre-processing



- Post-processing



- Learning with fairness constraints



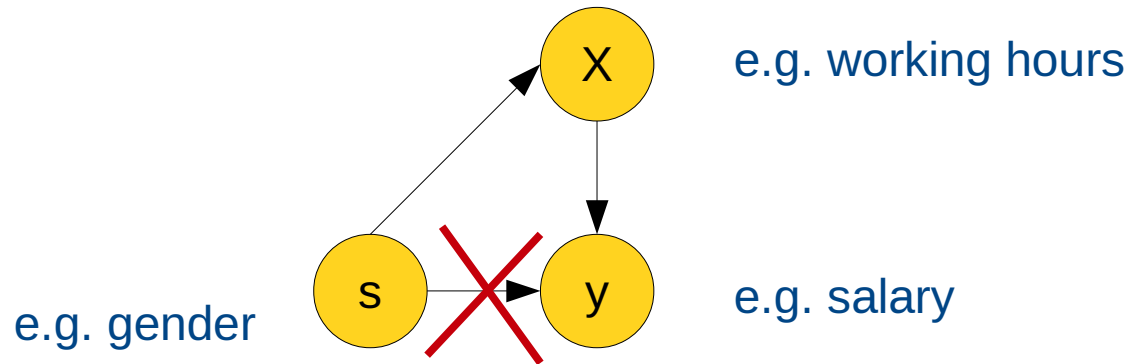
Algorithmic techniques

- Pre-processing
WORK WITH ANY ML
MODEL
- Post-processing
WORK ALMOST WITH ANY ML MODEL
- Learning with fairness constraints
EACH PAIR OF ML MODEL AND
FAIRNESS CRITERIA NEED A SPECIAL
ALGORITHMIC SOLUTION

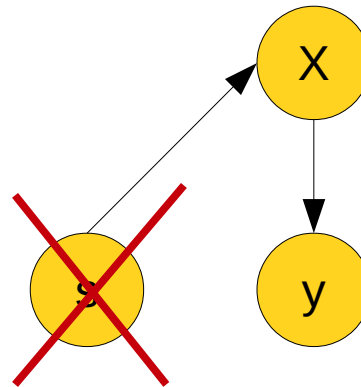
Algorithmic techniques in practice?

Pre-processing type 1

- Unlink s and y

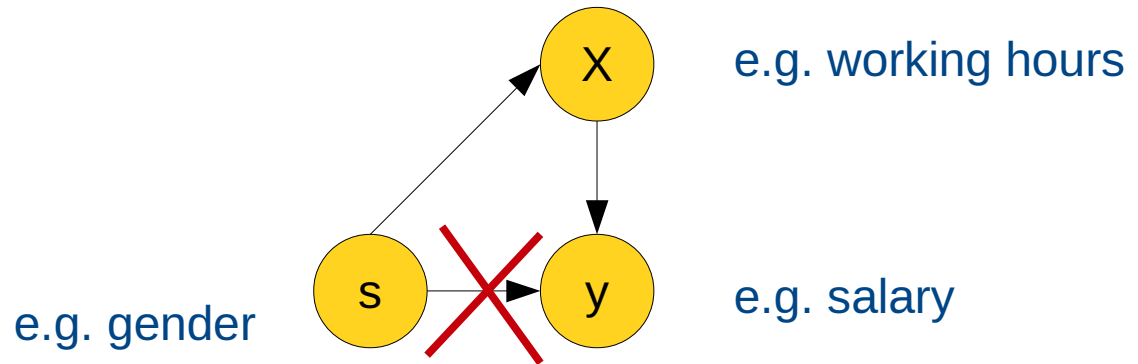


- Learn a regular model

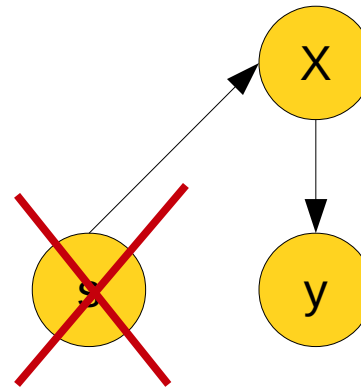


Pre-processing type 1

- Unlink s and y



- Learn a regular model



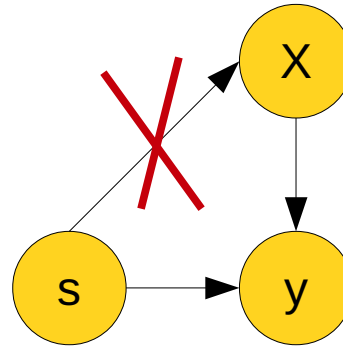
For example

Gender	Hours per week	Annual salary
Male	40h	45 000
Female	30h	25 000

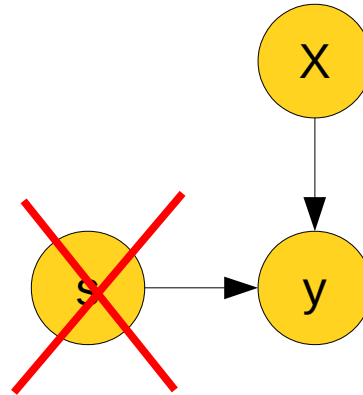
Gender	Hours per week	Annual salary
Male	40h	40 000 45 000
Female	30h	30 000 25 000

Pre-processing type 2

- Unlink s and X

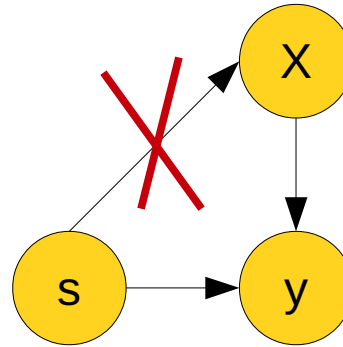


- Learn a regular model

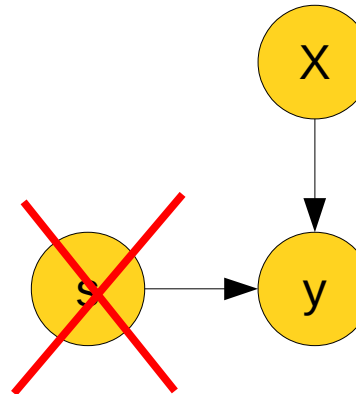


Pre-processing type 2

- Unlink s and X



- Learn a regular model



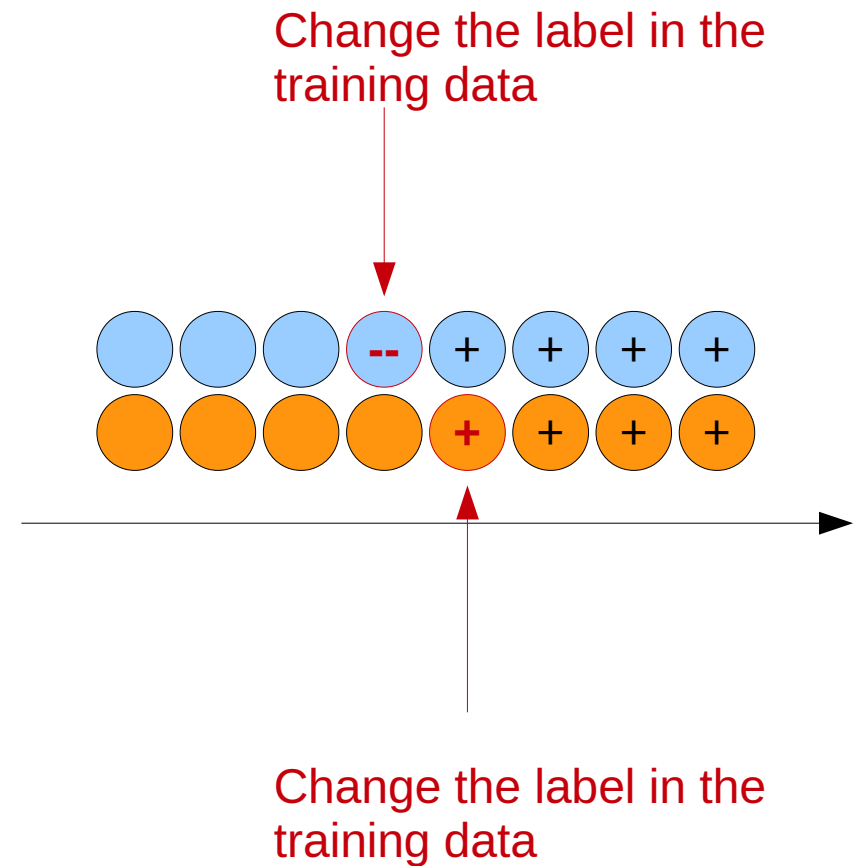
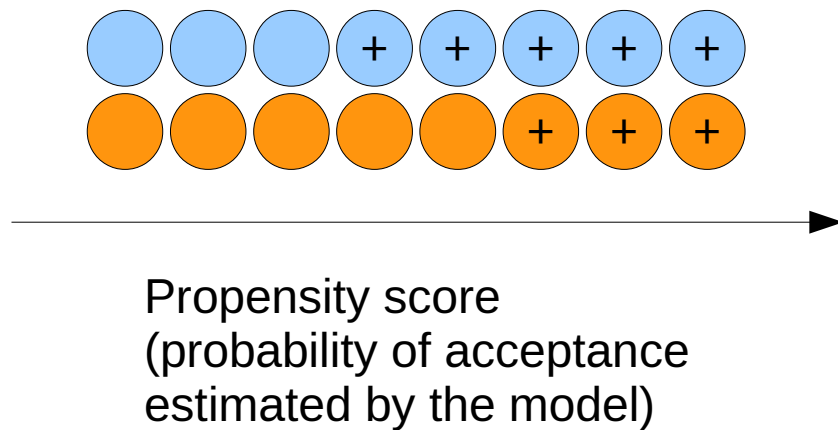
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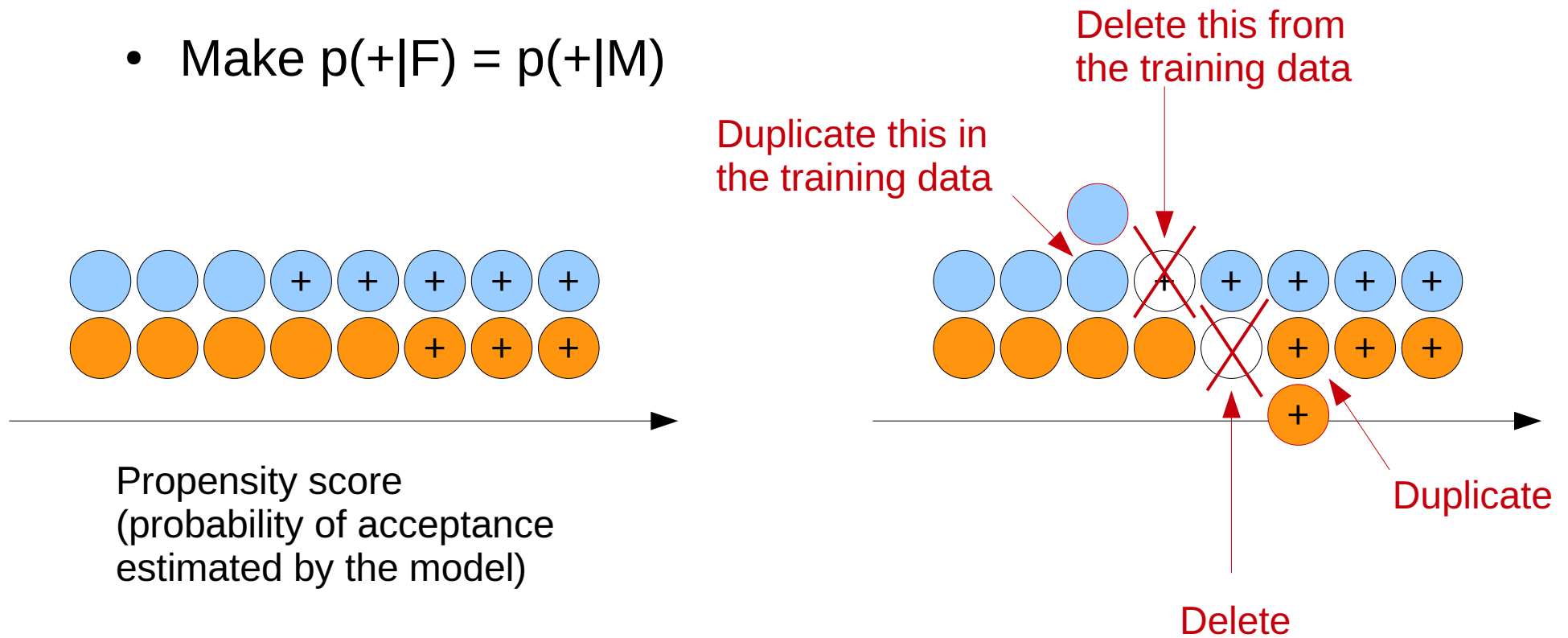
Massaging

- Unlink s and y
 - Make $p(+|F) = p(+|M)$



Preferential resampling

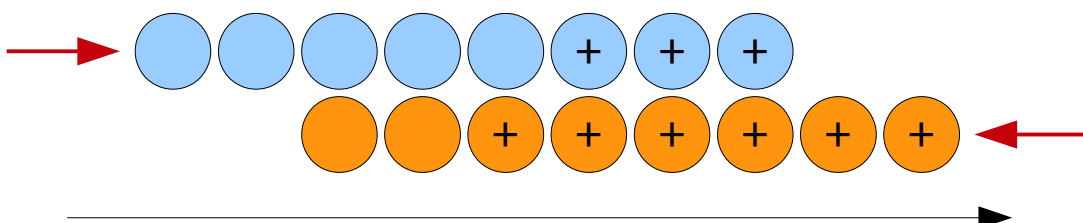
- Unlink s and y
 - Make $p(+|F) = p(+|M)$



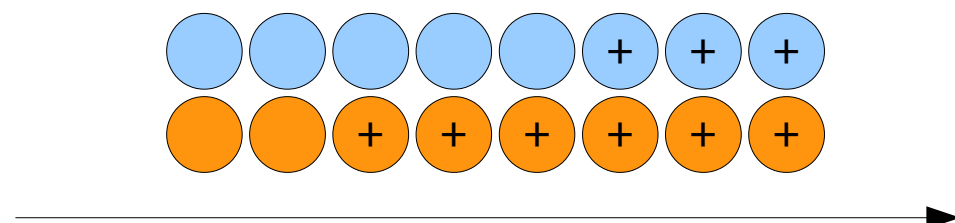
Combinatorial repair and variants thereof

- Unlink s and X
 - make $p(X|F) = p(X|M) = p(X)$

Modify X for each individual, either equally or in proportion to its distance to the median

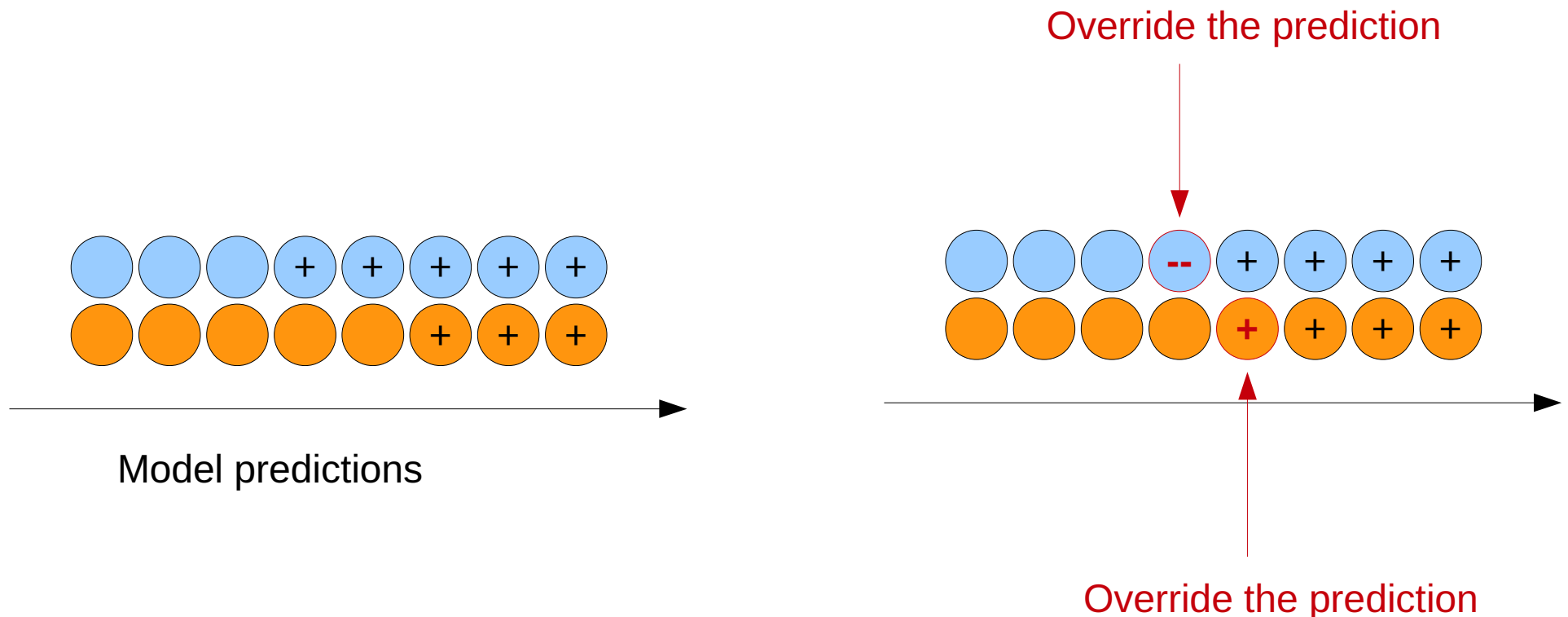


explanatory variable X ,
e.g. years of education



Post-processing of outputs

- Works in same way as massaging of inputs
 - Quota systems in hiring work this way, manual correction
 - How about online? Sliding window?



Post-processing the learned model

- Suppose historically salary is decided as

$$\text{salary} = 1000 + 100 \times \text{education} - 500 \times \text{ethnicity}$$

- Data scientists assumes

$$\text{salary} = b_0 + b_1 \times \text{education}$$

- Observes data

<i>education</i>	<i>ethnicity</i>	<i>salary</i>
1	1	600
2	1	700
3	1	800
4	1	900
10	1	1500

<i>education</i>	<i>ethnicity</i>	<i>salary</i>
1	0	1100
6	0	1600
7	0	1700
9	0	1900
10	0	2000

- Learns model

$$\text{salary} = 602 + 128 \times \text{education}$$

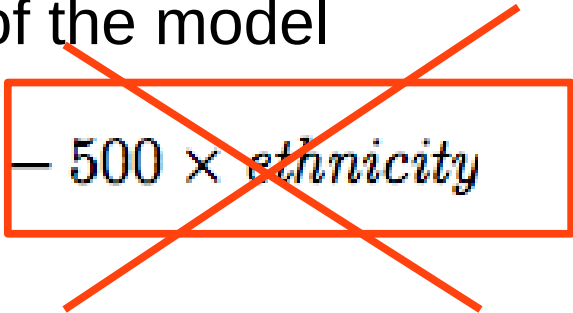
Lower base salary,
higher reward for education
the model punishes
ethnic minorities

Post-processing of the model

- Learn a model on the full dataset

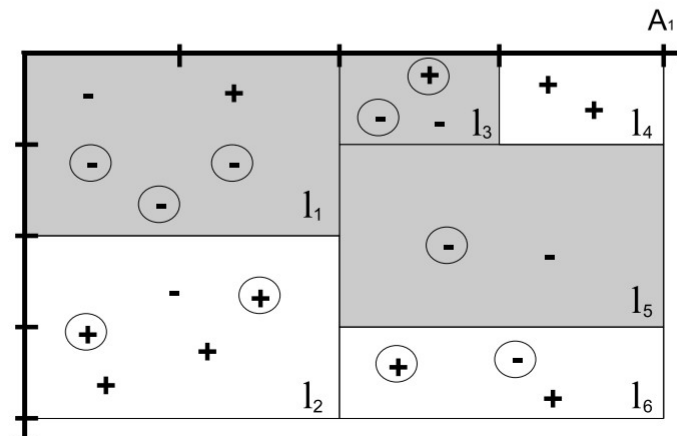
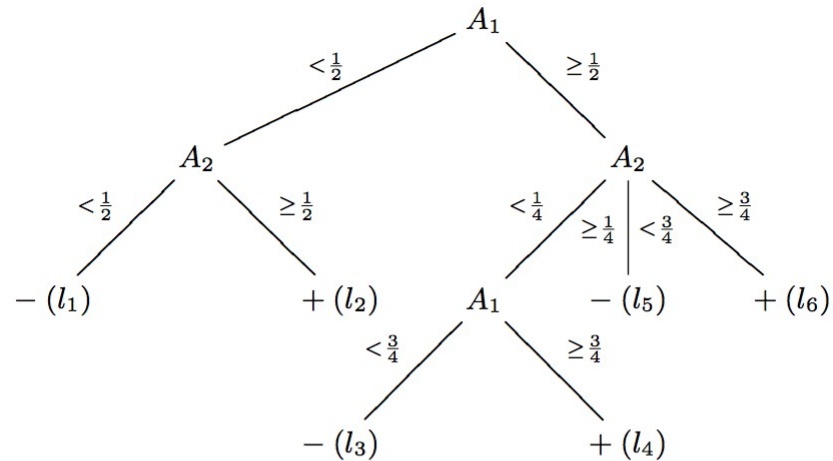
$$\textit{salary} = 1000 + 100 \times \textit{education} - 500 \times \textit{ethnicity}$$

- Remove the sensitive component of the model

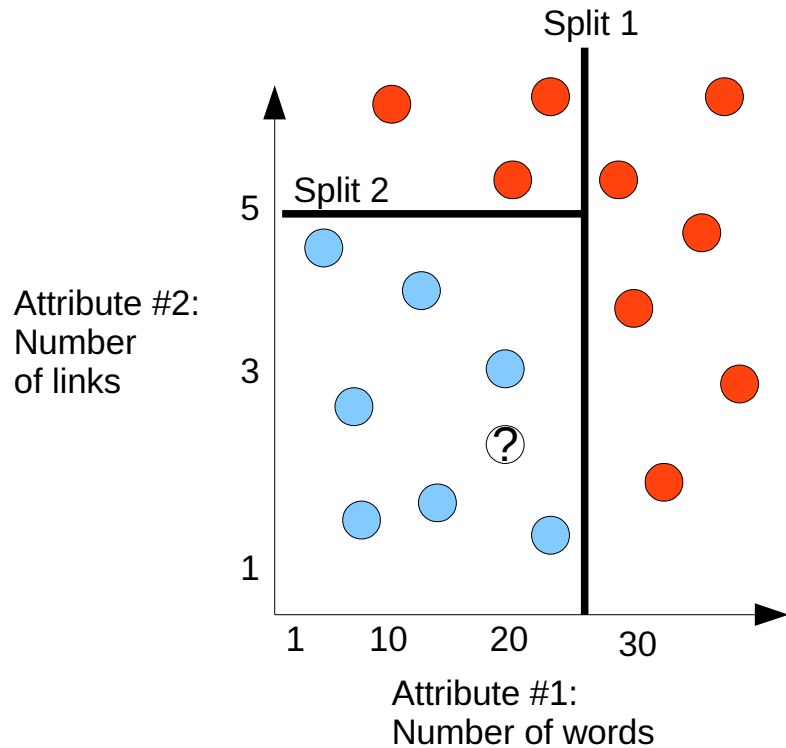

$$\textit{salary} = 1000 + 100 \times \textit{education} - 500 \times \textit{ethnicity}$$

Post-processing of the model

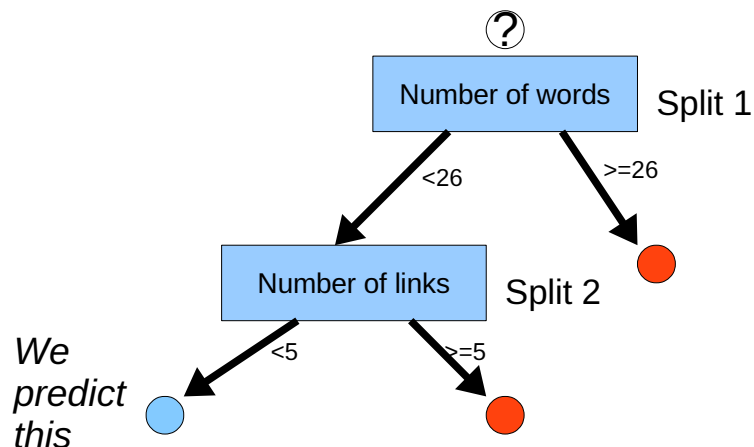
- Flip the labels of the tree leaves with the least damage to predictive accuracy until selected fairness metric is satisfied



Learning with constraints, decision tree



- Reminder: training a regular tree
 - A tree is constructed in divide and conquer manner
 - Calculate how much information about the label is provided by each attribute
 - Split at the most informative attribute
 - Continue until all nodes are pure (all labels are the same in a node), or pre-defined depth is reached
 - If the resulting tree is very large, *pruning* is used, i.e. some little informative nodes are cut



Learning with constraints, decision tree

Regular tree induction

$$IGC := H_{Class}(D) - \sum_{i=1}^k \frac{|D_i|}{|D|} H_{Class}(D_i)$$

Entropy wrt class label

Data subset
due to split

Discrimination-aware tree induction

$$IGS := H_B(D) - \sum_{i=1}^k \frac{|D_i|}{|D|} H_B(D_i)$$

Entropy wrt protected characteristic

Tree splits are decided on: $IGC - IGS$

Learning with constraints, logistic regression

model

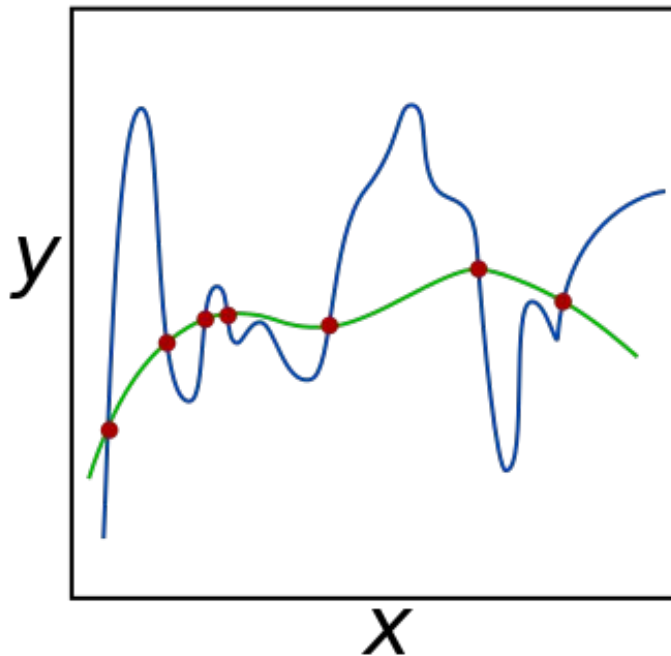
$$\mathcal{L}(\mathcal{D}; \Theta) = \sum_{(y_i, \mathbf{x}_i, s_i) \in \mathcal{D}} \ln \mathcal{M}[y_i | \mathbf{x}_i, s_i; \Theta].$$

Minimize:

$$-\mathcal{L}(\mathcal{D}; \Theta) + \eta R(\mathcal{D}, \Theta) + \frac{\lambda}{2} \|\Theta\|_2^2,$$

Fairness
regularizer

Standard
regularizer



Regular regression

Idea sketch: weights of explanatory variables that are strongly associated with sensitive variables should grow slower

Legal/ societal perspective on algorithmic sanitization techniques

- Pre-processing
 - Modify input data?
 - Resample input data?
- Post-processing
 - Modify models?
 - Modify outputs?
- Learning with constraints?

Legal/ societal perspective on algorithmic sanitization techniques

- Pre-processing

- Modify input data
- Resample input data



- Post-processing

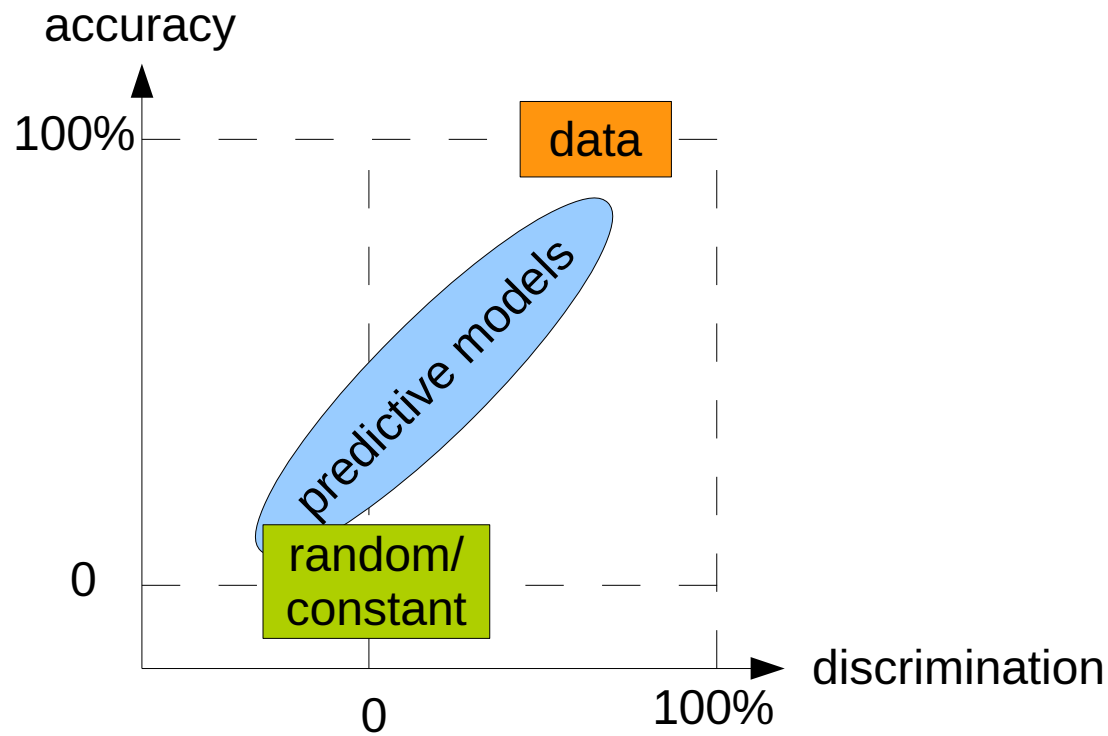
- Modify models
- Modify outputs



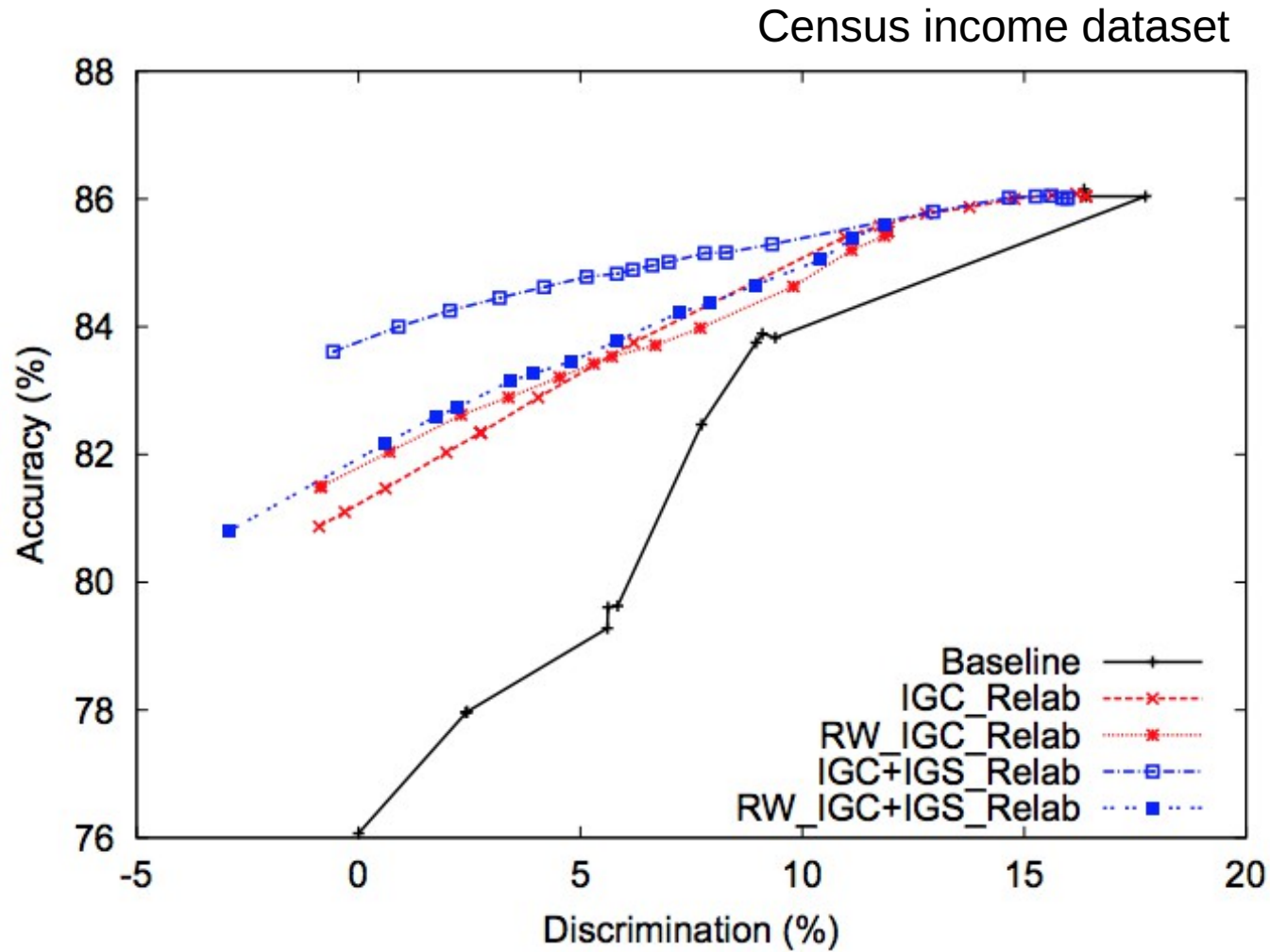
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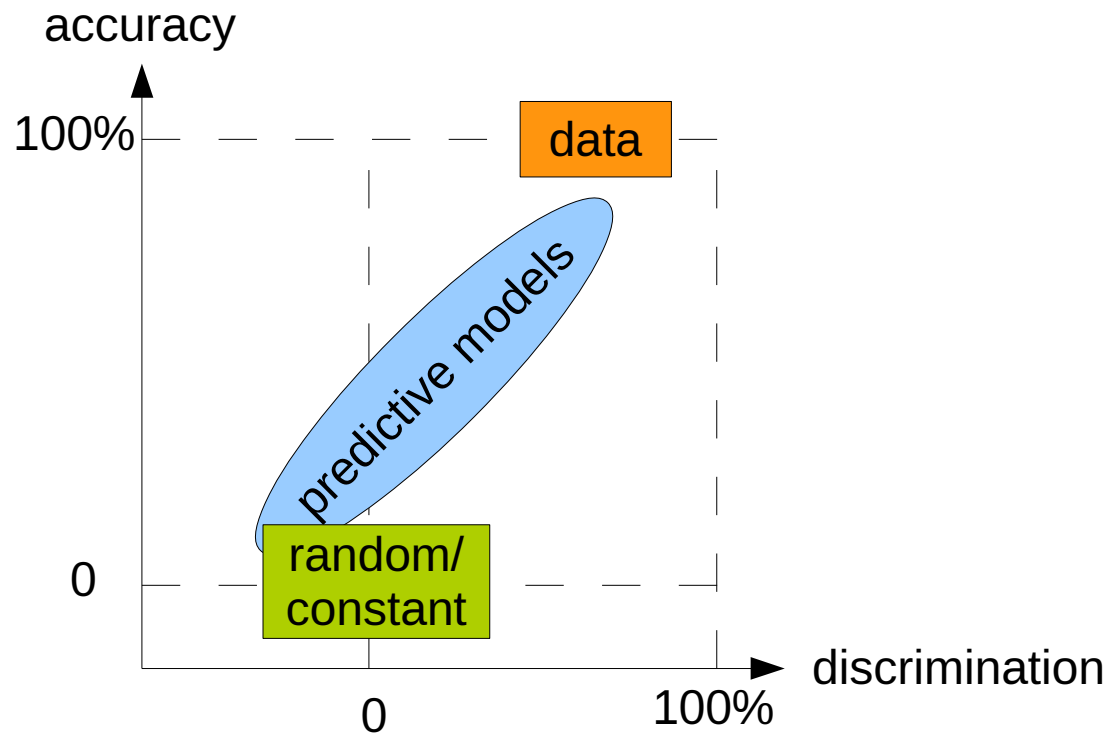
Accuracy-discrimination trade-off ??



Accuracy vs. discrimination



Accuracy-discrimination trade-off ??

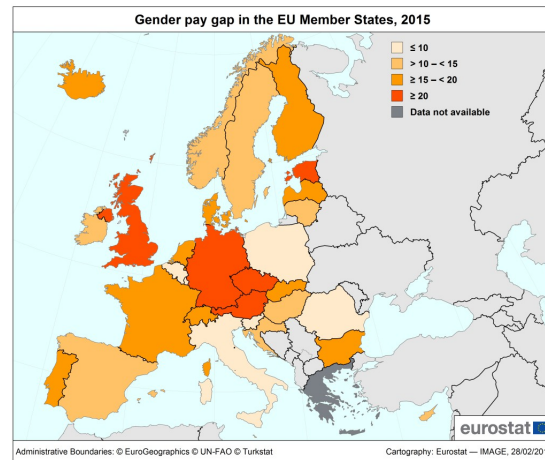


Shouldn't removing discrimination improve accuracy?

Why unbiased computational processes
can lead to discriminative decision procedures

Why unbiased computational processes can lead to discriminative decision procedures

- data is “incorrect”
 - due to biased decisions in the past



Why unbiased computational processes can lead to discriminative decision procedures

- Data is incomplete (e.g. omitted variable bias)



Image source:

https://www.syfy.com/sites/syfy/files/styles/syfy_episode_recap_full_breakpoints_theme_syfy_smartphone_1x/public/2018/03/futurama_408_hero.jpg

Why unbiased computational processes can lead to discriminative decision procedures

- Sampling procedure skews the data



Why unbiased computational processes can lead to discriminative decision procedures

- The world is changing over time

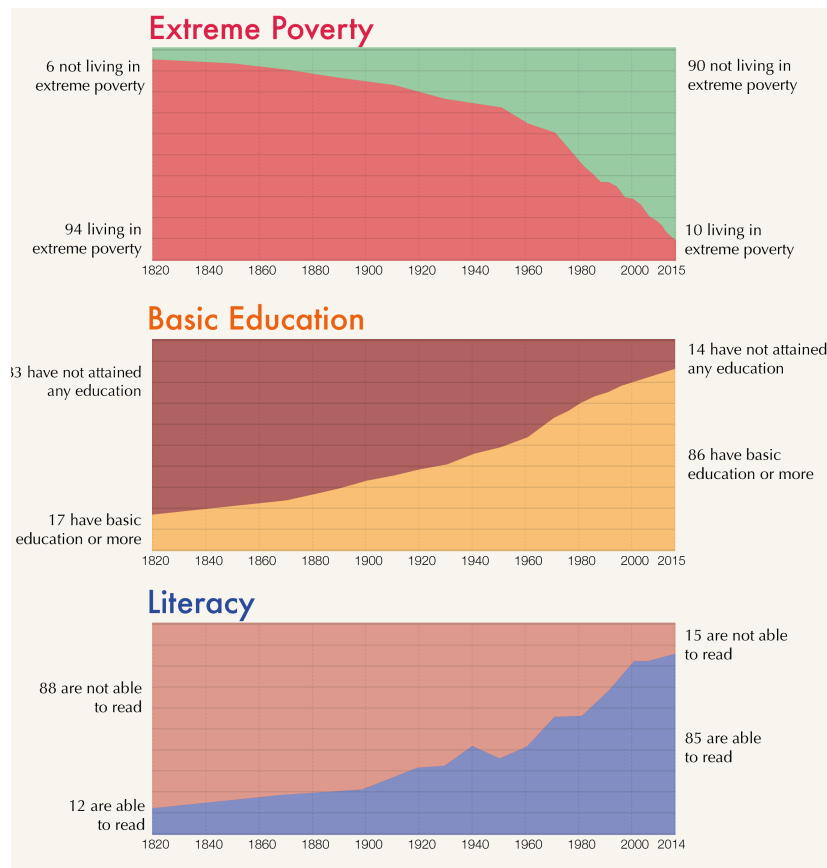


Image source:
<https://ourworldindata.org/a-history-of-global-living-conditions-in-5-charts/>
<https://ritholtz.com/2017/04/world-100-people-last-two-centuries/>

Why unbiased computational processes can lead to discriminative decision procedures

- Data is “incorrect”
- Data is incomplete
- Sampling procedure skews the data
- The world is changing over time



Easy to fix when effort is applied

Why unbiased computational processes can lead to discriminative decision procedures

- Data is “incorrect”
- Data is incomplete
- Sampling procedure skews the data
- The world is changing over time

Easy to fix when effort is applied

difficult



ARTIFICIAL INTELLIGENCE

Inside Amsterdam's high-stakes experiment to create fair welfare AI

The Dutch city thought it could break a decade-long trend of implementing discriminatory algorithms. Its failure raises the question: can these programs ever be fair?

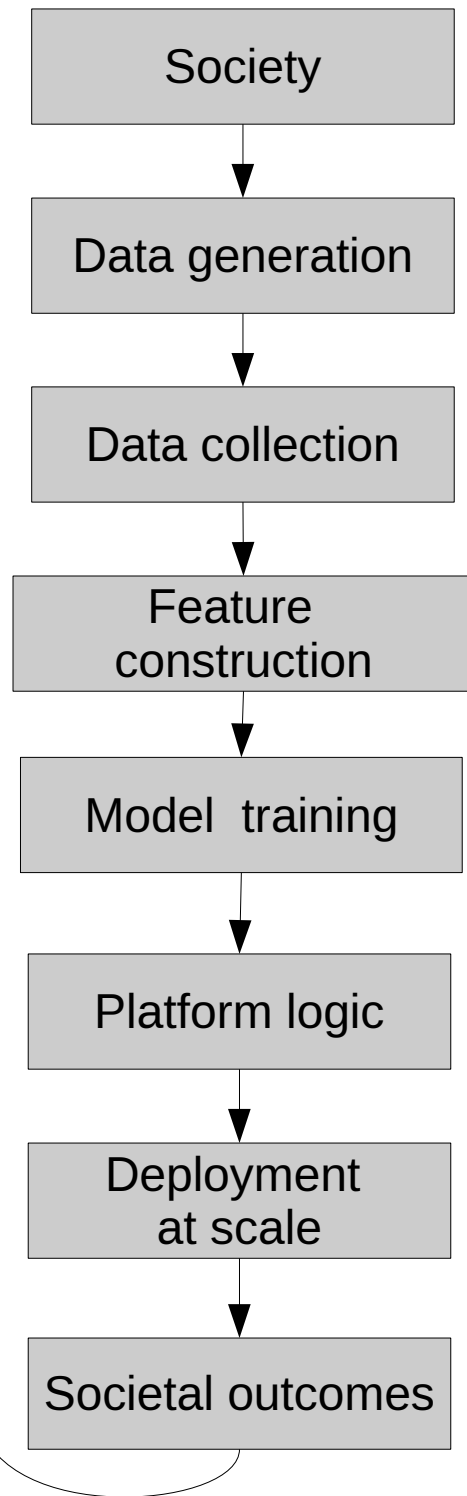
Case: A “Fair” Welfare AI – Amsterdam (2025)

- Goal: detect welfare fraud more fairly than humans
- Actions done
 - Interpretable models
 - Bias testing and audits
 - No sensitive attributes
 - Consultants and “responsible AI” guidelines

Case: A “Fair” Welfare AI – Amsterdam (2025)

- Goal: detect welfare fraud more fairly than humans
- Actions done
 - Interpretable models
 - Bias testing and audits
 - No sensitive attributes
 - Consultants and “responsible AI” guidelines
- Pilot testing
 - Disproportionally flagged vulnerable groups
 - No clear improvement over human case workers
 - In some dimensions performed worse
- Outcome: never deployed after years of development

Why did the Amsterdam case fail?



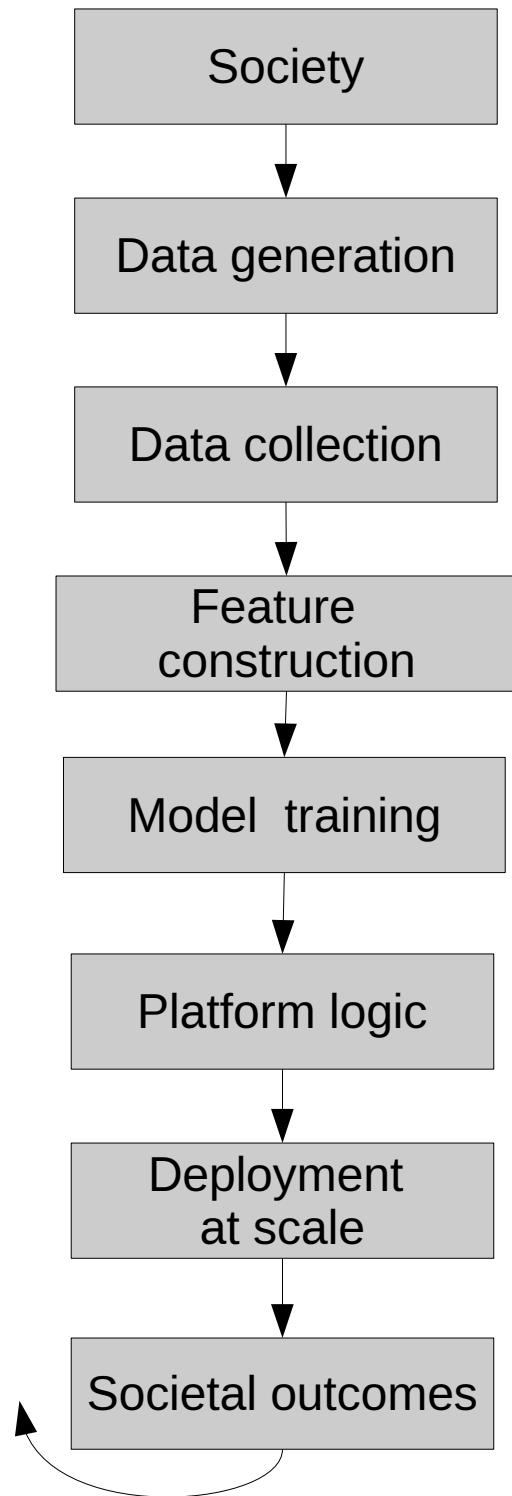
Description	Feature importance ranked
Percentage participation in welfare schemes in previous year	1
Days since last moved house	2
No-contact or no-reply appointments in previous year	3
Sum of assets	4
Days since last shift	5
Co-occupants?	6
Received benefit in previous year?	7
Average 'points docked' for violating benefits rules	8
Applied for benefit in previous year?	9
Average of gross income	10
Address in Amsterdam?	11
Single or partnered?	12
No-show appointments in previous year	13
Active addresses	14
Sum of gross income	15

Source: <https://www.lighthousereports.com/methodology/amsterdam-fairness/>

Why did the Amsterdam case fail?

Explainable Boosting Machine

- Tree-based model
- From Microsoft



Amsterdam case, model building

TrainTest – human flagged cases ~3000

Pre-pilot – cases flagged by humans and the model ~1300

Pilot – cases flagged by the model ~300

Fairness criterion: equal false positive rates for groups



Problem 1: Non-identifiability of fairness measures

Flagged for investigation,
Investigated,
true fraud confirmed

Flagged for investigation,
Investigated,
no fraud found

		True outcome	
		Positive	Negative
Predicted class	Positive	TP true positives	FP false positives
	Negative	FN false negatives	TN true negatives

Not investigated,
true outcome unknown

Problem 1: Non-identifiability of fairness measures

Observability under uniform sampling

		True outcome	
		Positive	Negative
Predicted class	Positive	TP true positives	FP false positives
	Negative	FN false negatives	TN true negatives

Observability under selective sampling with **extra audit**

		True outcome	
		Positive	Negative
Predicted class	Positive	TP	FP
	Negative	FN*	TN*

unknown

real FN real TN

Problem 1: Non-identifiability of fairness measures

Measure	Statistic	Identified?
<i>Measures on positively predicted cases</i>		
Statistical parity	$(FP + TP)/N$	•
Predictive Parity, Precision	$TP/(FP + TP)$	✓
False Discovery Rate	$FP/(FP + TP)$	✓
<i>Measures on negatives</i>		
Predictive Equality, False Positive Rate	$FP/(FP + TN)$	×
True Negative Rate, Specificity	$TN/(FP + TN)$	×
False Omission Rate	$FN/(FN + TN)$	×
Treatment Equality	FN/FP	×
<i>Measures on true outcomes</i>		
Equal Opportunity, True Positive Rate	$TP/(TP + FN)$	×
False Negative Rate	$FN/(TP + FN)$	×
Equalised Odds, Separation	TPR & FPR	×
Accuracy	$(TP + TN)/N$	×
<i>Other measures</i>		
Calibration / Sufficiency	$P(y = 1 \mid f, a)$	×
FP Group	FP/N	•

Problem 1: Non-identifiability of fairness measures

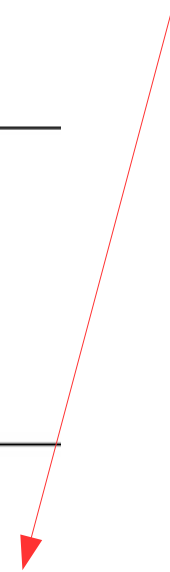
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harm to innocents
unidentified

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Calibration / Sufficiency	$P(y = 1 f, a)$	×
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harm to innocents
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The original model was biased against non-Dutch nationals

Needles show the percentage chance of the selected group being falsely flagged. Percentage chance is in relation to the reference group (e.g. men if women is selected).

Bias against

Non-Western Nationality

Women

Non-Dutch Nationality

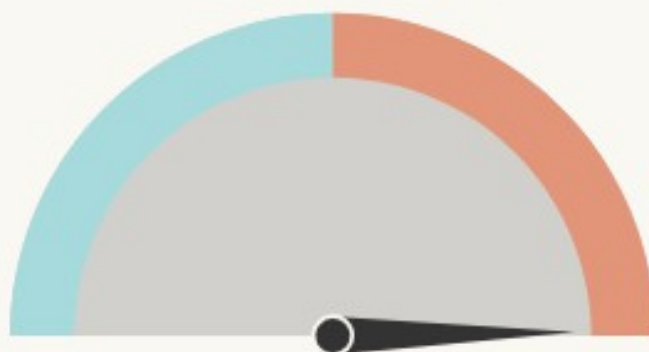
Analogue Process

-34%



Initial Model

99%



After reweighing, the model wrongly flagged Western and non-Western applicants at equal rates

Needles show the percentage chance of the selected group being falsely flagged. Percentage chance is in relation to the reference group (e.g. men if women is selected).

Bias against

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Analogue Process

-34%



Initial Model

99%



Reweighed Model

0%



THE DEBIASED MODEL IS DEPLOYED IN A PILOT

Amsterdam's debiased model showed bias against Dutch nationals and women in the pilot

Needles show the percentage chance of the selected group being falsely flagged. Percentage chance is in relation to the reference group (e.g. men if women is selected).

Bias against

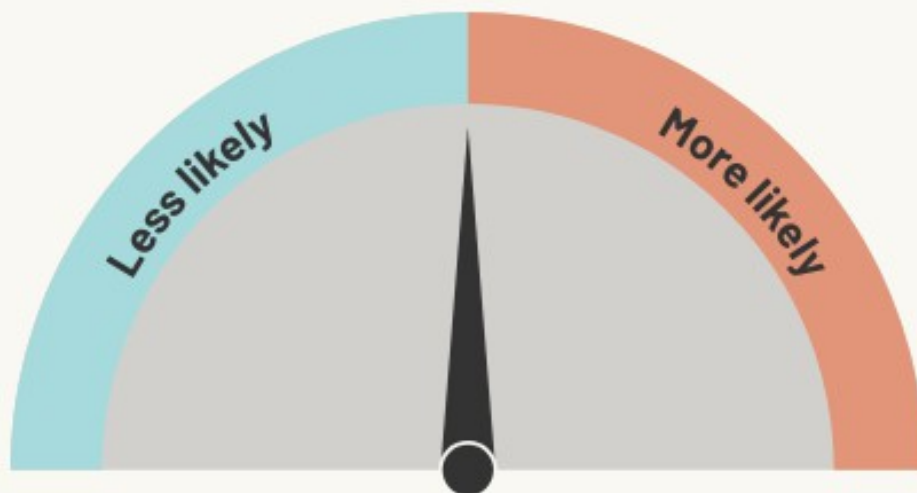
Non-Western Nationality

Women

Non-Dutch Nationality

Reweighed Model

0%

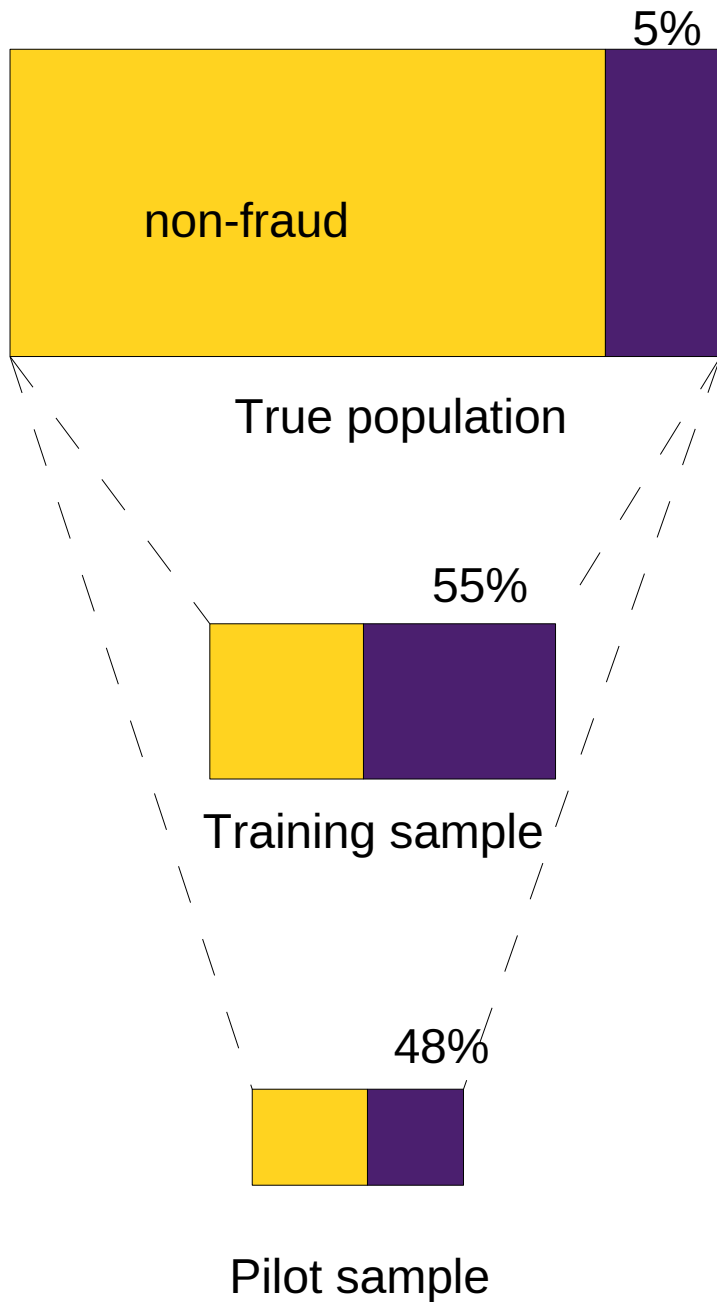


Reweighed Model (Pilot)

-38%



What has happened?



Unidentifiability of fairness measures

Observability under uniform sampling

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		Positive	Negative
Predicted class	Positive	TP true positives	FP false positives
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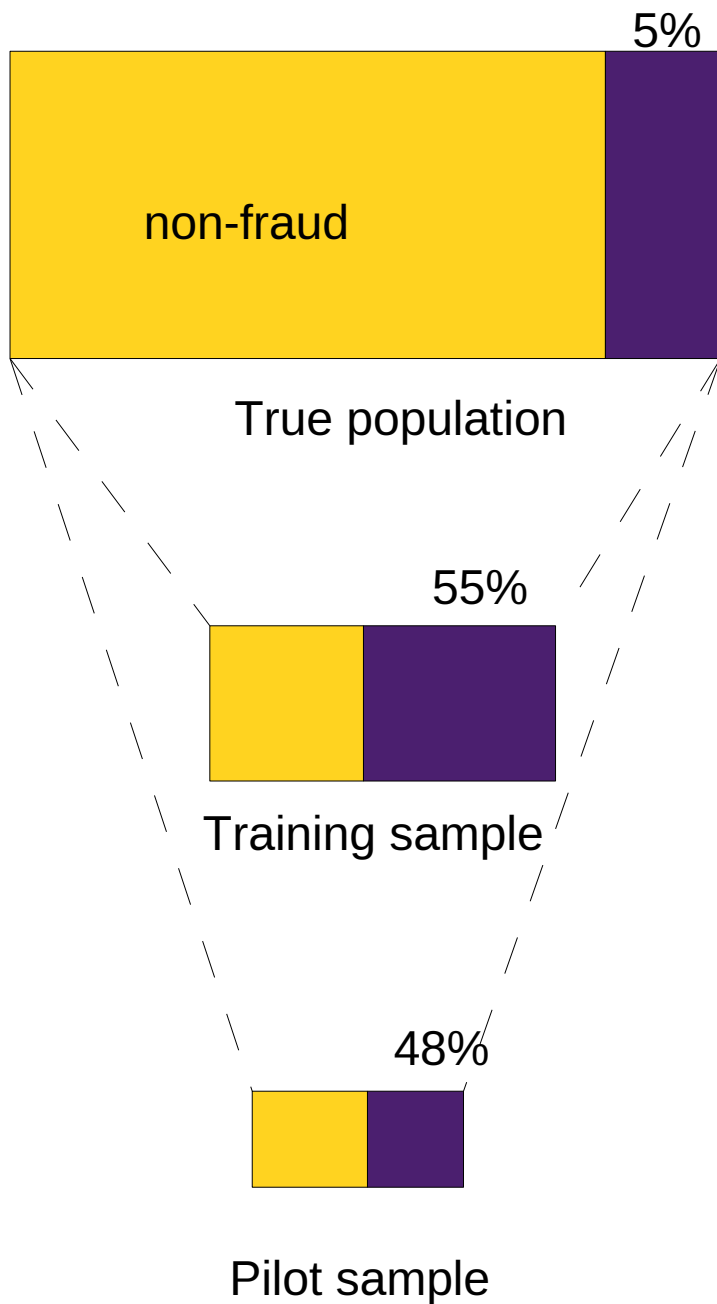
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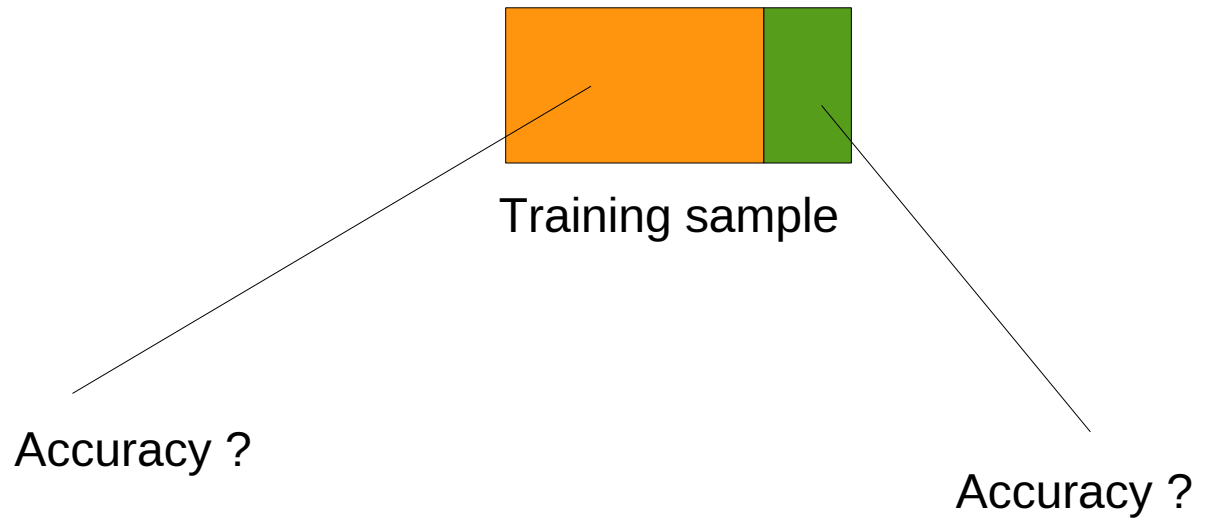
unknown

real FN real TN

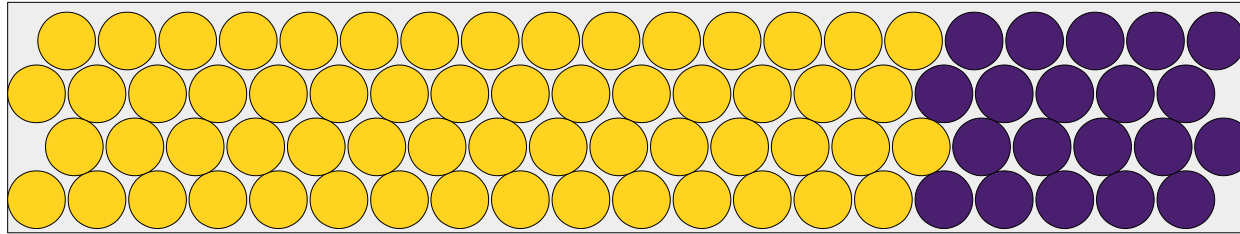
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What has happened?

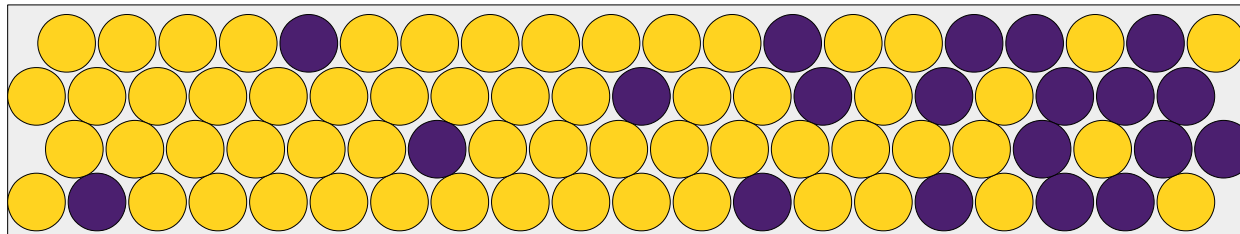


Accuracy ~ ranking

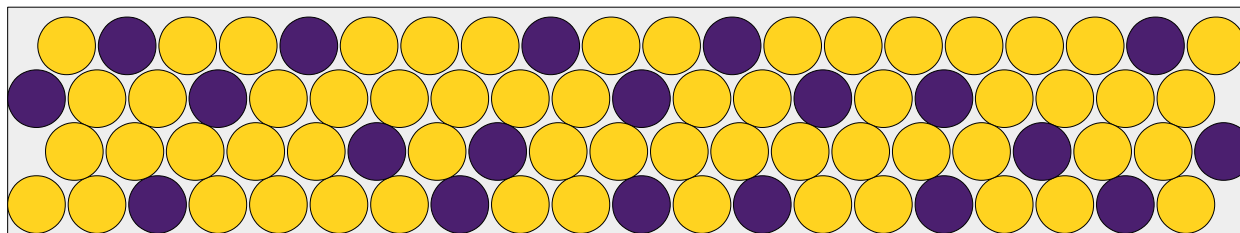


Perfect classifier

Model scores

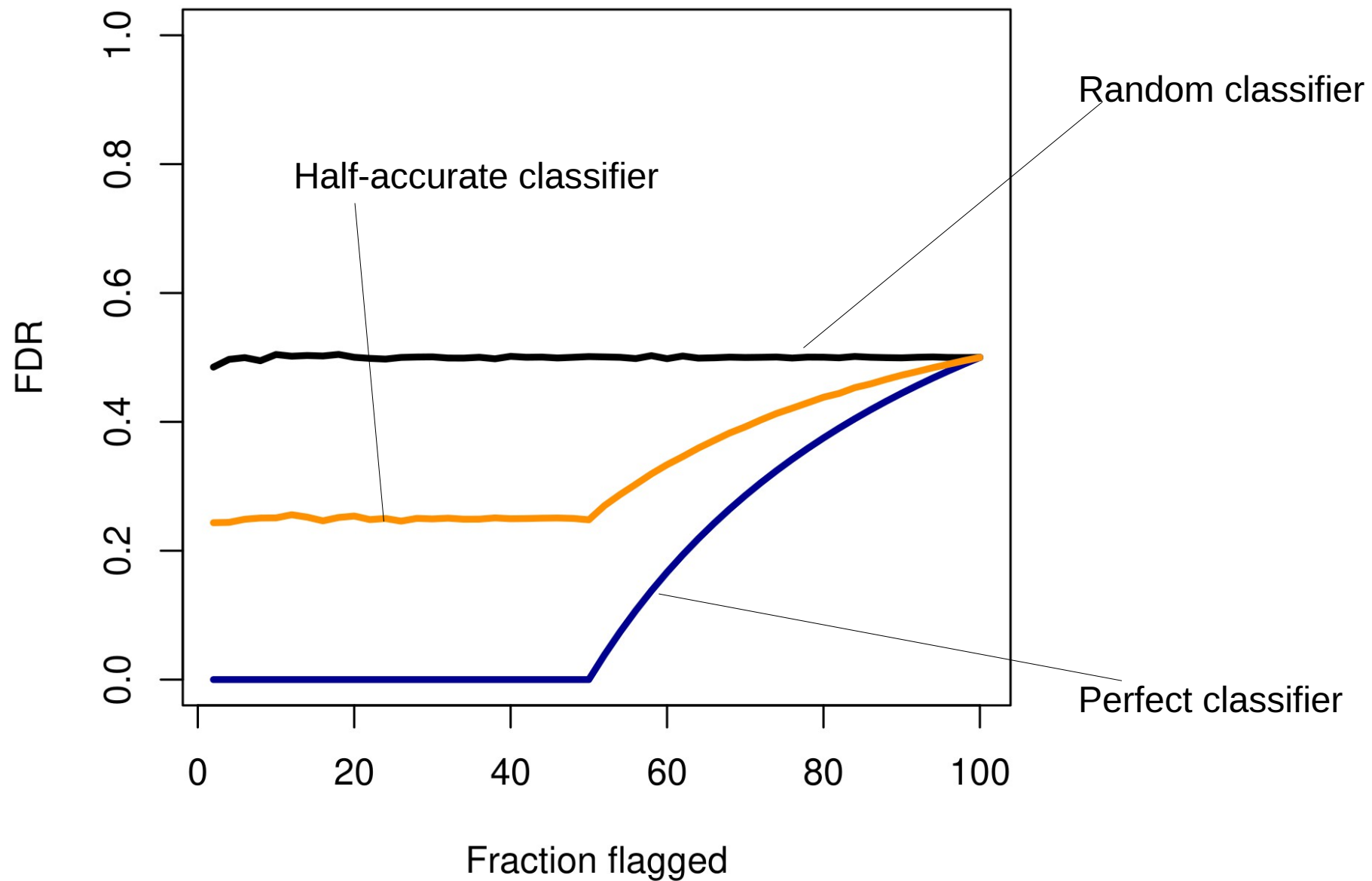


Half accurate classifier

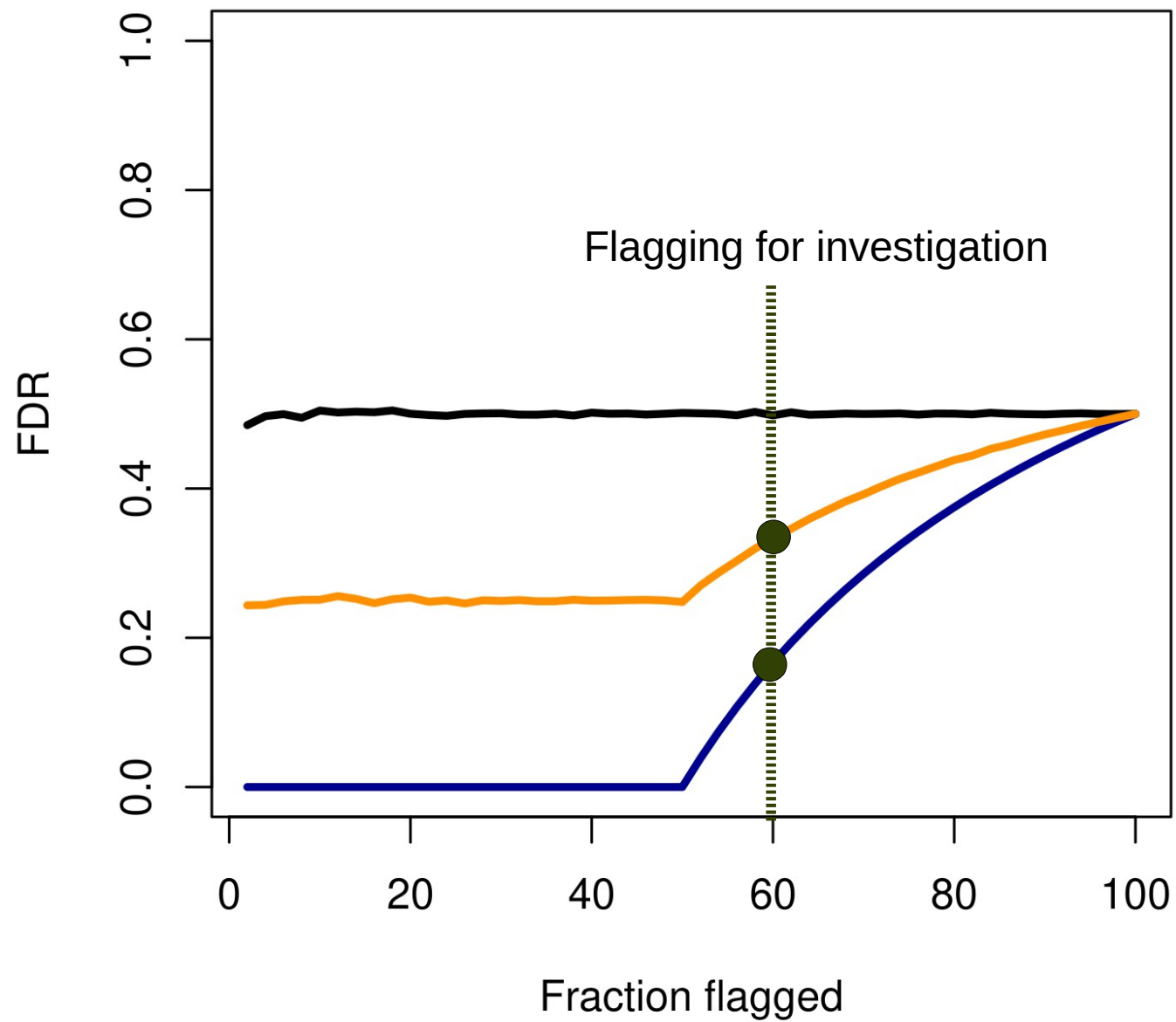


Random classifier

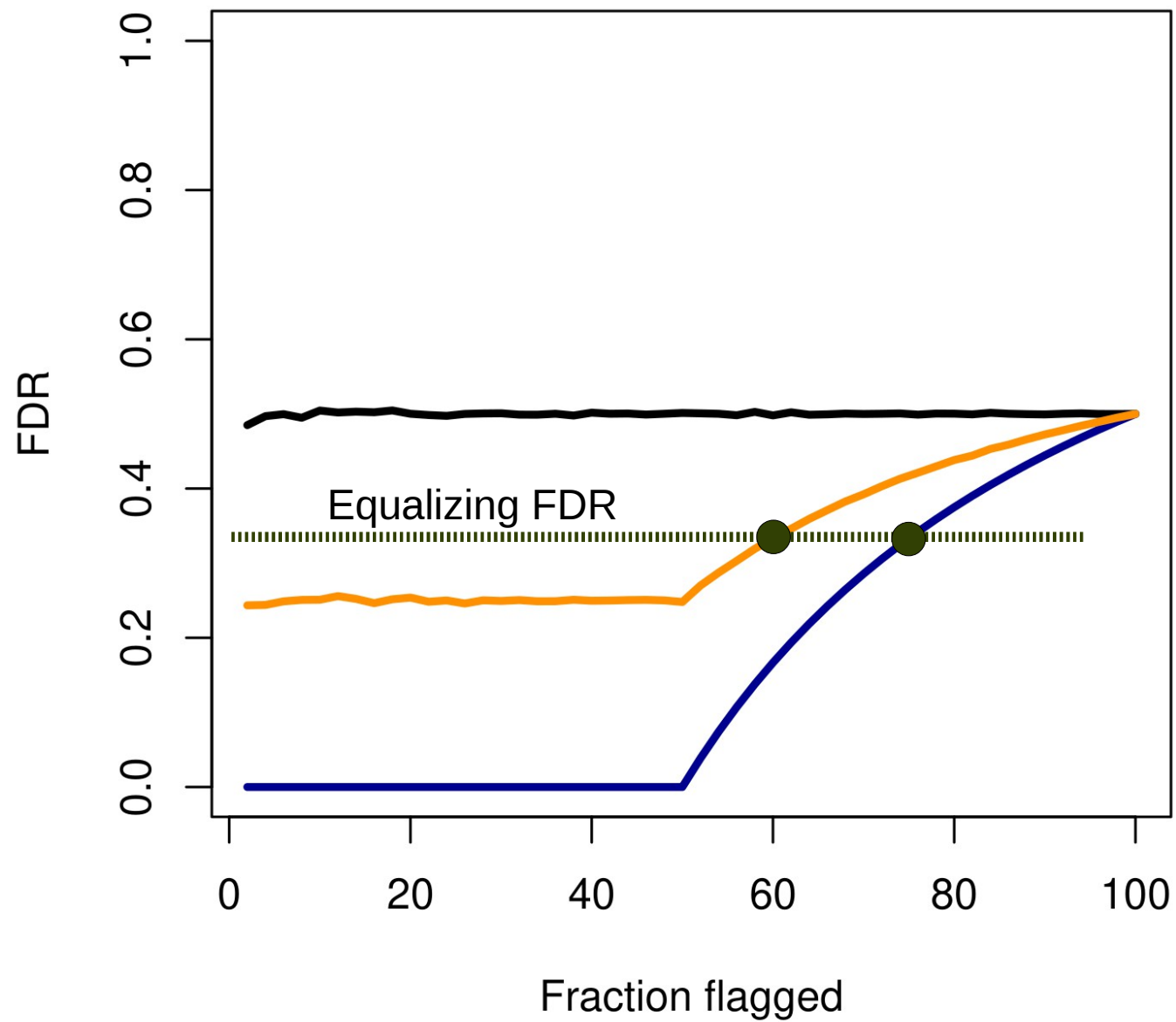
False discovery rates



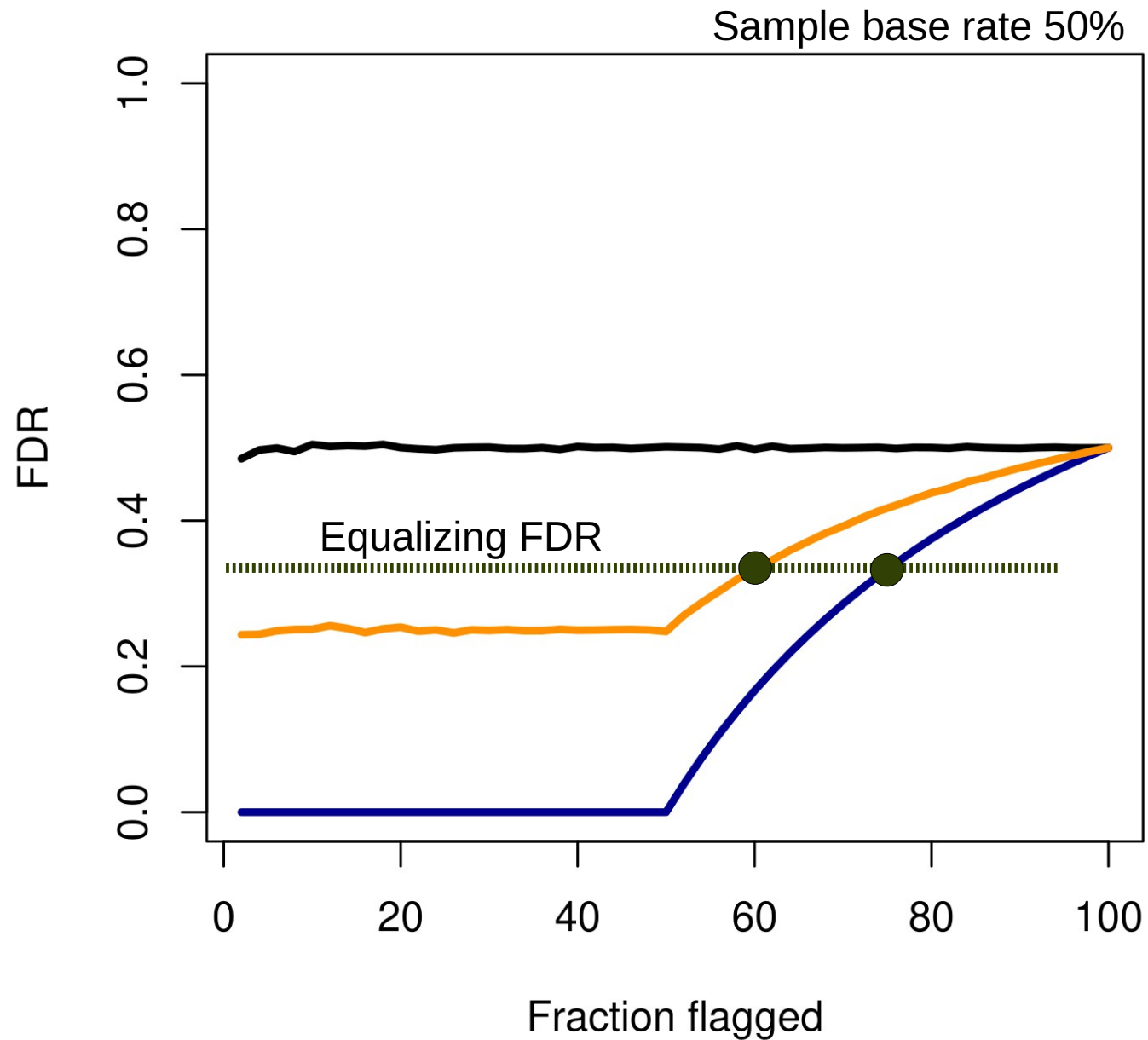
False discovery rates



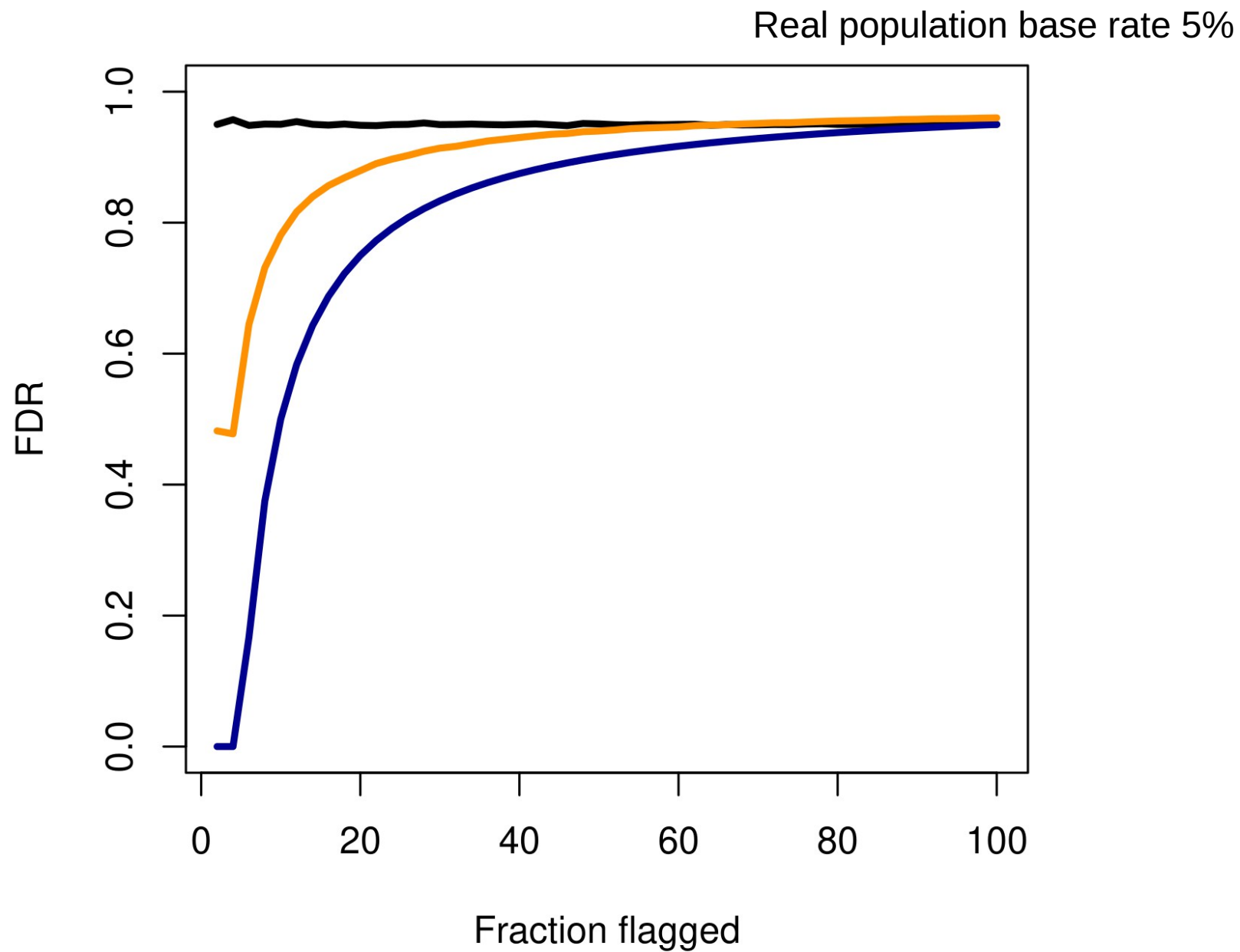
False discovery rates



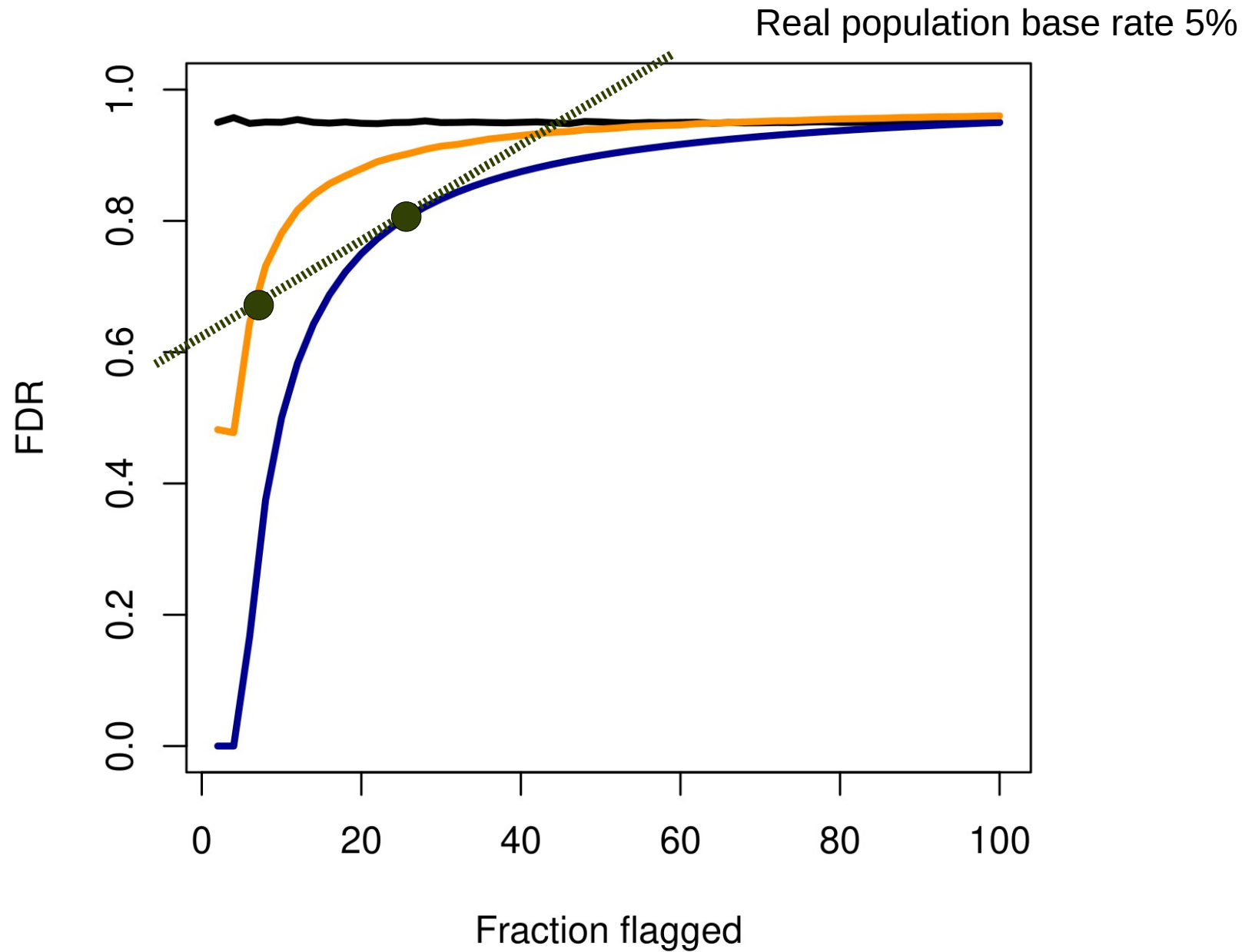
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False discovery rates



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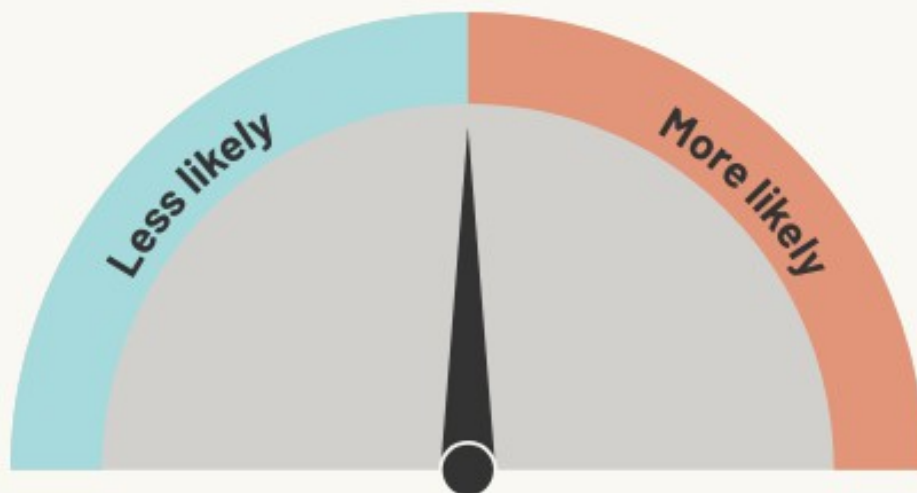
Non-Western Nationality

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Non-Dutch Nationality

Reweighed Model

0%



Reweighed Model (Pilot)

-38%



Large scale online platform decisions



Content

- Showing posts, ranking of search results, recommendations

Access to

- credit approval, pricing, jobs, housing

Governance

- content removal, account suspension, safety interventions

Opportunity

- ad delivery, visibility of creators/sellers, matchmaking, dating

Why do we still keep being surprised by algorithmic discrimination?

- Not lack of intent
- Not lack of methods
- Not lack of regulation

Why do we still keep being surprised by algorithmic discrimination?

- We fix models, not systems
- We mistake technical properties for moral guarantees
- We delegate decisions without relocating responsibility
 - accountability remains human, agency quietly moves to algorithms
- We evaluate statically, systems operate dynamically

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Scientific way forward?

- Yes — if we change the unit of analysis
 - From models → systems
 - From static metrics → dynamic evaluation
 - From optimization of accuracy → capturing of failure modes and boundary conditions
- Make fairness a system property, not a model property
 - Causal models to bound the harm, not to prove fairness